5. (*a*) What is the range of error in rounding? 2

(*b*) What are quantization steps? Tabulate the quantization steps in 3-bit binary representation of all the three fixed point formats. 7

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(*c*) The output of an of A/D converter is applied to a digital filter with the system function

$$H(z) = \frac{0.45z}{z - 0.72}$$

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Find the output noise power for the digital filter, when the input signal is quantized to 7 bits of range input signal is -1 to 1. 7

(*d*) Discuss the effect of coefficient quantization on pole location of the following IIR system, when it is realized in direct form –1. Assume o word length of 4 bits through truncation $H(z) = \frac{1}{1 - 0.7z^{-1} + 0.12z^{-2}}$ 7

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325655(25)

BE (6th Semester)
Examination, April - May, 2016

(New Scheme)

Digital Signal Processing

Time Allowed : 3 hours *Maximum Marks :* 80
Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

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1. (*a*) What is a twiddle factor? 2

(*b*) Perform (*i*) Linear convolution

(*ii*) Circular convolution, to determine the response of a system $h(n) = \{2, \underset{\uparrow}{2}, 1, 1\}$ to an input $x(n) = \{1, \underset{\uparrow}{2}, 0, 1\}$. 7

(*c*) (*i*) Find the DFT of the sequence $x(n) = \{1, -1, 2, -2\}$. 3

(*ii*) Find the 1DFT of $X(K) = \{4, 2, 0, 4\}$. 4

(*d*) Compute the 1DFT of $X(K) = \{12, 0, 0, 0, 4, 0, 0, 0\}$ using FFT algorithm. 7

2. (*a*) Mention the condition that are to be satisfied by an impulse response of an FIR system to have a linear phase. 2

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(*b*) If $x(n) = \{\underset{\uparrow}{1}, 3, 5, 2, 3, 6, 1, 2\}$, compute signal in frequency domain using DCT algorithm. 7

(*c*) The impulse response of a low-pass filter is given by 7

$$h_d(n) = \begin{cases} \dfrac{1}{\pi(n-\zeta)} \sin \omega_c (n-\zeta) & \text{for } n \neq \zeta \\ \dfrac{\omega_c}{\pi} & \text{for } n = \zeta \end{cases}$$

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Design a low-pass FIR filter for $N = 5$ and $\omega_c = 0.5$ radian using Hamming window.

(*d*) Write a short note on manimelly flat FIR filters. 7

3. (*a*) Give two requirements for an analog filter to be causal and stable. 2

(*b*) Determine $H_a(s)$ for a Butterworh filter satisfying the following constraints in digital domain : 7

$$\sqrt{0.5} \leqslant |H(e^{j\omega})| \leqslant 1 \qquad 0 \leqslant \omega \leqslant \frac{\pi}{2}$$

$$|H(e^{j\omega})| \leqslant 0.2 \qquad \frac{3\pi}{4} \leqslant \omega \leqslant \pi$$

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with $T = 1$ sec. [Use reletion $\Omega = \dfrac{\omega}{T}$]

(*c*) The system function of an analog filter is excpressed as

$H_a(s) = \dfrac{2}{s(s+2)}$. Find the corresponding $H(3)$ using the impulse invarient method for a sampling frequency of 4 samples per second. 7

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(*d*) What is frequency warping? How will it arise? 7

4. (*a*) What is the advantage of direct form II structure over direct form I structure? 2

(*b*) (*i*) Realise the FIR system given with minimum number of multiplies

$$H(z) = 1 + \frac{B}{15} z^{-1} + \frac{17}{15} z^{-2} + \frac{8}{15} z^{-3} + z^{-4}$$ 3

(*ii*) Realize the FIR system given in cascade form.

$$H(z) = \left(1 + \frac{1}{3} z^{-1} + z^{-2}\right) + \left(1 + \frac{1}{5} z^{-1} + z^{-2}\right)$$ 4

(*c*) An LTI system is described by the equation

$y(n) + 2y(n-1) - y(n-2) = x(n)$

Determine the parallel realization structure of the system. 7

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(*d*) Compare between FIR and IIR system. 7